SAGE

A tradition-faithful retrieval-augmented pipeline for moral & spiritual reflection

Sacred Alchemy & Guidance Engine

SAGE

Sacred Alchemy & Guidance Engine

A tradition-faithful retrieval-augmented pipeline for moral & spiritual reflection

Catherine M Smith · University of Virginia · DS 5002 Final Project

huggingface.co/smithykitty/SAGE-Guidance-Engine

github.com/smithykitty/SAGE

Bible | Qur'an | Gita | Dhammapada | Tao Te Ching

EXECUTIVE SUMMARY

Retrieval — not the model — is what makes it work

THE PROBLEM

- Asked for scripture, general LLMs fabricate verses
- They inherit a documented Western-Christian skew
- They offer no citable provenance for what they claim

SAGE keeps scripture outside the model — it retrieves real passages, filters them to the seeker's tradition, and cites each by name.

WHAT I FOUND

19% → 98%

RGB accuracy, closed-book → with retrieval (same generator)

0% → 62%

Citation grounding on SAGE's own testing split

SAGE runs on the weakest base model tested, yet beats every comparison once retrieval is added.

Why a general LLM is a poor spiritual guide

People bring moral quandaries — forgiveness, duty, grief, honesty — to their traditions. A general model answers from parametric memory, and in this domain a confident misattribution can distort a person's relationship with their own faith.

Fabricates verses

Invents plausible-sounding scripture that was never written.

Western-Christian skew

Over-represents dominant traditions; under-serves the rest.

No provenance

Cannot point to a real, checkable source for its claims.

SAGE'S APPROACH

Keep scripture external → **Retrieve real passages** → **Filter to the seeker's tradition** → **Cite each by name**

DATA

38,467 verses across five traditions + a 100-case gold set

RETRIEVAL CORPUS · public-domain scripture

Christian	King James Bible	31,102
Islamic	Qur'an — Saheeh Intl. English	6,236
Hindu	Bhagavad Gita — Sivananda	701
Buddhist	Dhammapada — Buddharakkhita	347
Taoist	Tao Te Ching — Legge	81

One verse per chunk, tagged with its tradition and canonical reference (e.g. Matthew 18:21).

38,467
verse chunks

100
gold test cases

30
seeker profiles

5 / 10
traditions built

GOLD TEST SET

30 synthetic profiles (age, gender, seeker→disciple) × moral quandaries. Each case carries the ideal passages it should retrieve and an ideal reference answer.

METHODOLOGY

A tradition-filtered RAG pipeline — no fine-tuning

Quandary	Embed	Retrieve	Generate	Reflection
seeker's question	sentence-transformer	top-5 · cosine · tradition-filtered	Qwen2.5-7B + 5 exemplars	cited · non-prescriptive

Why RAG over fine-tuning: the task needs verifiable, citable scripture, and the knowledge is a large public-domain corpus best kept external and updatable (Lewis et al., 2020).

ENCODER COMPARISON · Recall@5 on the test prompts

0.015	0.025	0.028
`all-MiniLM-L6-v2`	`bge-small-en-v1.5`	`all-mpnet-base-v2`
384-d baseline	*384-d, retrieval-tuned*	*768-d · WINNER*

The weakest base model wins once you add retrieval

Model	RGB	MultiHop	Grounding
SAGE (Qwen2.5-7B + RAG)	**98%**	**64%**	**62%**
Qwen2.5-7B — base, closed-book	19%	47%	0%
Llama-3.1-8B — comparison	28%	44%	—
Mistral-7B — comparison	36%	58%	—

n = 100, seed 5002 · SAGE row supplies gold passages; base row is closed-book

+79 pts

RGB lift from retrieval alone

Mistral beats base Qwen closed-book (36% vs 19%) — yet SAGE, built on Qwen, beats them all once retrieval is added.

→ the gain is the pipeline, not the model.

Load the pipeline; reflect, don't rule

LOAD & RUN

```
from transformers import AutoModelForCausalLM, AutoTokenizer
from sentence_transformers import SentenceTransformer

tok = AutoTokenizer.from_pretrained(
        "Qwen/Qwen2.5-7B-Instruct")
gen = AutoModelForCausalLM.from_pretrained(
        "Qwen/Qwen2.5-7B-Instruct", device_map="auto")
emb = SentenceTransformer("BAAI/bge-small-en-v1.5")

# retrieve tradition-filtered passages, then generate
print(sage("A friend asked me to cover for
     something they did. What should I do?",
     ["christian"]))
```

INTENDED

- Personal reflection on a moral question
- Comparative study across traditions
- A demo of tradition-faithful RAG

NOT INTENDED

- A source of religious authority
- A substitute for clergy or a teacher
- A mental-health or crisis resource

Grounded in, and citing, the retrieved passages

PROMPT FORMAT

```
SYSTEM: You are SAGE, a non-prescriptive
spiritual companion. Reflect grounded ONLY
in the retrieved passages, cite each by name,
and end with "Sources: <ref>; <ref>".

USER:
QUANDARY
  <the seeker's moral quandary>

RETRIEVED PASSAGES
  - Matthew 18:21: Then came Peter...
  - Colossians 3:13: Forbearing one another...
```

EXPECTED OUTPUT

A 250–400 word, non-prescriptive reflection that surfaces tension and convergence across the cited passages, closing with a machine-readable Sources line.

"Forgiveness, in the passages you've been given, is framed less as a single act than as a posture returned to over time. When Peter asks how many times... [~300 words]

Sources: Matthew 18:21; Colossians 3:13"

Retrieval quality is the ceiling

Retrieval is the bottleneck

Best encoder reaches Recall@5 ≈ 0.03. The generator grounds 62% of citations, but surfaces the ideal verse only ~7% of the time.

Five of ten traditions

Jewish, Norse, Aboriginal, Gandhian & Teresan sources are oral or in-copyright — 34% of test cases have nothing to retrieve.

Register gap

Modern first-person quandaries vs. archaic KJV / classical translations depresses embedding similarity.

Evaluation scope

Synthetic profiles; gold answers inherit the teacher model's style; RAGTruth faithfulness not yet run.

The pipeline matters more than the base model's parametric knowledge.

SAGE grounds a 7B model in real, tradition-filtered scripture — turning confident fabrication into checkable citation.

Next step: building an engaging UI/UX.

Explore the repository

huggingface.co/smithykitty/SAGE-Guidance-Engine

github.com/smithykitty/SAGE